Forecasting German Electricity Demand: A Comparison of Benchmark, SARIMAX, Feature-Based and Neural Forecasting Models

Mathujan Sivananthan - 24087340

GitHub Repo Link:
https://github.com/sMathujan/Time-Series-Modelling-Case-Study---24087340

1. Introduction

This report forecasts the electricity load in Germany with consideration of a wide variety of simple benchmarks to more and more complicated models, such as SARIMAX, feature-based tree models and neural networks. The research questions the models address are: (1) which models can meaningfully outperform a seasonal naive model; (2) how to prevent data leakage in features; and (3) why certain hyperparameter values are selected, with the differencing order being one example. In addition, we investigate if adding temperature covariates increases the accuracy of the forecast, examine the compromise between being easy to interpret and being more complex, and finally suggest a model that could be used in an operation. The main argument of this analysis is that it is very hard to beat a trivial seasonal naive benchmark fairly. However, on a real forecasting assignment on a fixed origin with a two-year-ahead forecast horizon, no complex model has any meaning when compared with it, mainly because of the structural change that occurred at the time of the COVID-19 pandemic.

2. Data and Preprocessing

The main dataset employed in this study is the Open Power System Data (OPSD) actual load data of Germany for 1 hour from January 2015 to October 2020. Data were cleaned, and missing data were removed, and the hourly load in Megawatts (MW) was converted to a weekly mean in Gigawatts (GW). This led to 301 weekly observations, of which 197 were assigned for training, and 104 observations were assigned for testing. The key issue here is the aggregation of the data to a weekly frequency, in order to remove the dominant daily and weekend cycles, and reveal a more pronounced annual seasonal signal. A strictly chronological train/test split was enforced to prevent any data leakage. In the case of the Long Short-Term Memory (LSTM) model, the *MinMaxScaler* was applied only to the training period of the hours, and then it was applied to the whole series. Furthermore, all the lag and rolling features were created from the shift operation of the tree-based models (e.g., lag_1 = shift(1), drop_first(52) for lag_52), such that the features on week *t* only contain data until week *t-1*. It is important to note that the temperatures recorded during the test period represent future weather, and thus the temperatures used are, in this way, look-ahead data, rather than true operational ones, which makes the models used conditional forecasts.

3. Exploratory Analysis

This exploratory data analysis included visualising the raw series, performing seasonal and trend decomposition using Loess (STL), and examining the autocorrelation function (ACF) and partial autocorrelation function (PACF). The results of the honesty and critical tests on the level series were stationary, as indicated by the Dickey-Fuller test (p=0.001) and the KPSS test (p=0.10). The conclusion of this finding is interpreted as a strong deterministic annual seasonality as opposed to a stochastic unit root. This vital insight will give direction to the subsequent discussion of the suitable differencing orders for the SARIMA models.

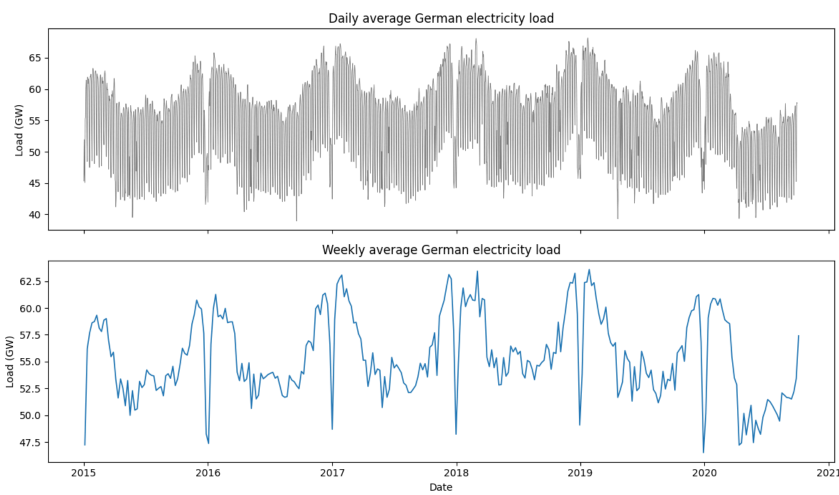

Figure 1: Weekly average and Daily average German electricity load

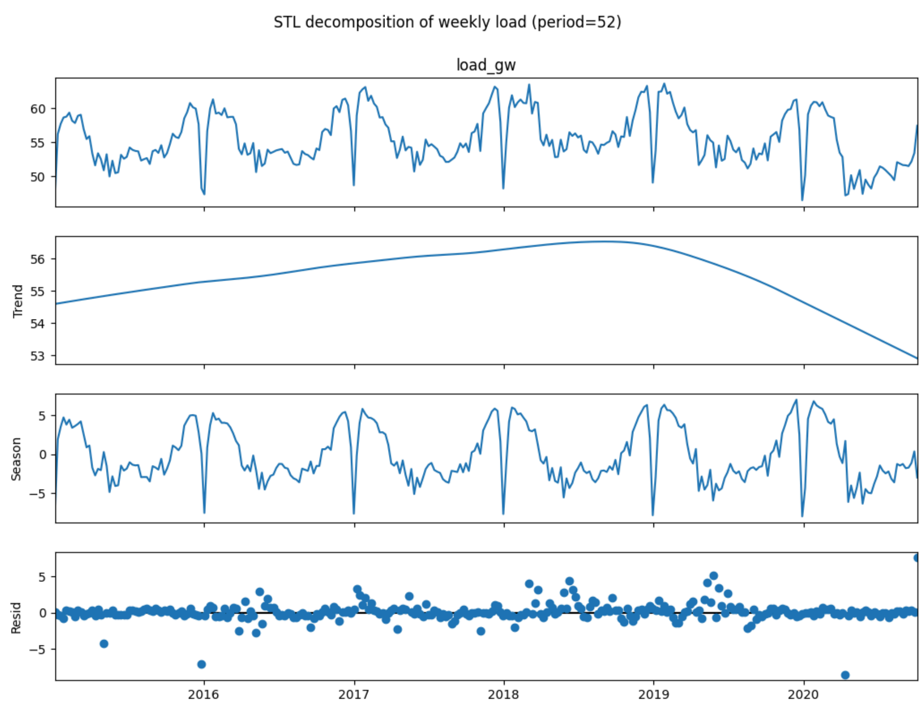

Figure 2: STL decomposition of weekly load

4. Forecasting Methods

Several model families were compared to standard benchmarks in the forecasting pipeline. A seasonal length of $s=52$ was selected for the SARIMAX model due to the presence of an annual cycle as seen in the ACF plot with a spike at lag 52 and the STL decomposition. To eliminate this annual mean, an airline-style seasonal differencing ($D=1$) was used. The initial tests of stationarity indicated that the non-seasonal differencing may not be required ($d=0$), but the AIC grid search favoured a non-seasonal differencing of $d=1$. Adding $d=1$ onto $D=1$ for a stationary level series is a mild level of over-differencing, which plausibly increases the bias of the model's extrapolation (by +2.80 GW) but helps the in-sample information criteria. The models based on features (Random Forest and Gradient Boosting Machines) were organised into a recursive multi-step forecaster. Engineered features were used as inputs to the models, with shift-based lags. In the forecasting step, the models recursively injected their forecasts as lags, instead of real future load data. Finally, an LSTM neural network was implemented for shorter time horizons. The ability of LSTMs to model complex sequences of time without much manual feature engineering is well documented in the field of load forecasting (Hochreiter & Schmidhuber, 1997; Kong et al., 2019; Marino et al., 2016; Bouktif et al., 2018). This network was trained, however, for rolling forecasts over a week ahead and not set for a fixed two-year time horizon due to architectural requirements.

5. Evaluation Design

The main evaluation task is at a fixed origin of 104 weeks. The models each provide a forecast at the end of the training period, and they each predict the forecast 2 years into the future. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Scaled Error (MASE), and Bias were used for evaluating models. The MASE model serves as a reference benchmark model that is anchored with the in-sample model seasonal naive.

An important imbalance to highlight is that the LSTM model is predicting a week-ahead forecast that is rolled across the test window, re-anchoring on the latest actuals. This is a much simpler task than the fixed-origin 2-year forecast which the other models perform, so it cannot be fairly or directly compared to the others.

6. Results

The comparative accuracy of the test window of 104 weeks is displayed in Table 1. First, the Random Forest model with its RMSE of 2.970 GW and second, the LSTM model with its RMSE of 2.155 GW, outperform the baseline model for the seasonality with its RMSE of 3.007 GW. The 1% difference in the performance of the Random Forest is, however, not statistically significant and devoid of meaningful value. But, 72% of the feature importance is based on lag_52, indicating that this model was able to learn the seasonal naive benchmark better compared to any meaningful improvement over it.

The deficit of the LSTM has been 28%, which seems a large number, but is rather confused and confused by its weekly time frame. On a "like for like" basis, the seasonal naive model is the best. Highly positive demand forecasts (+1.69 to +2.80 GW) in the Bias column are most robust for the models that are fixed at a particular origin. The seasonal naive model is very competitive because it simply copies the previous season, rather than projecting misguided past seasons onto the unobserved future.

Model	RMSE	MAE	MASE	Bias
LSTM (week-ahead, rolled)*	2.155	1.657	1.238	+0.53
Feature model — Random Forest	2.97	2.212	1.604	+1.69
Seasonal naive (benchmark)	3.007	2.319	1.732	+1.73
Feature model — Gradient Boosting	3.265	2.47	1.792	+1.82
SARIMAX + temperature	3.553	2.815	2.103	+2.52
SARIMA (2,1,6)(0,1,1,52)	3.762	3.08	2.301	+2.80
Mean	4.397	3.789	2.831	+0.48
Naive	4.459	3.783	2.827	-0.88
Drift	5.118	4.34	3.243	+1.01

Table 1: Weekly forecast accuracy on the 104-week test window (GW). Sorted by RMSE.

* LSTM is a week-ahead forecast rolled across the test window (re-anchored on recent actuals), NOT a fixed-origin 2-year forecast — not directly comparable.

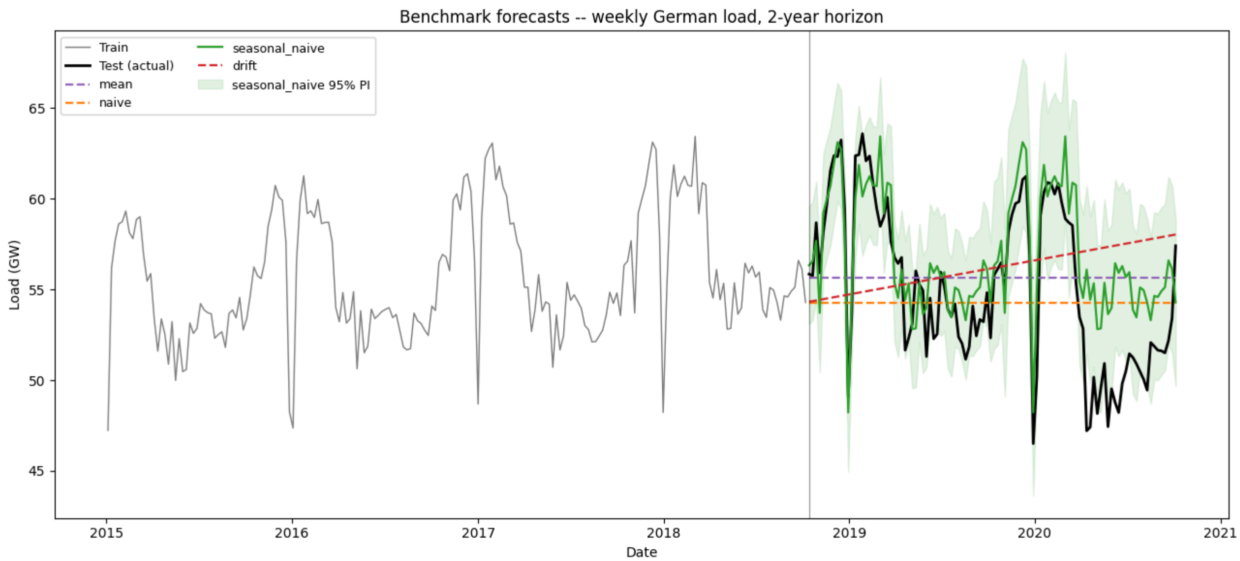

Figure 3: Seasonal naive (Benchmark) forecasts

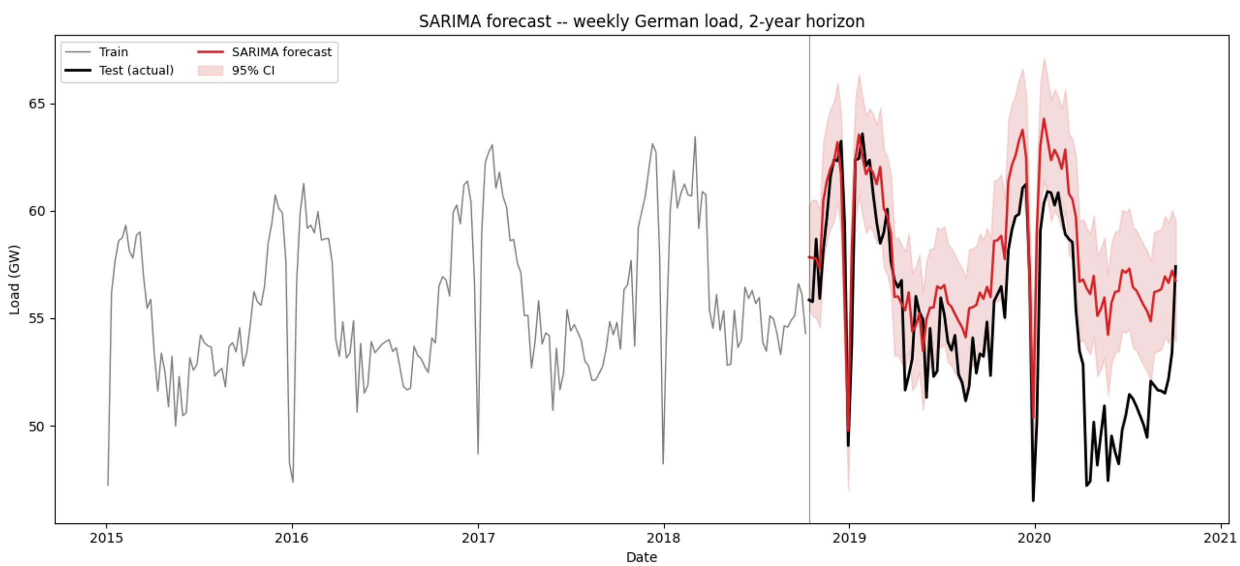

Figure 4: SARIMA forecast

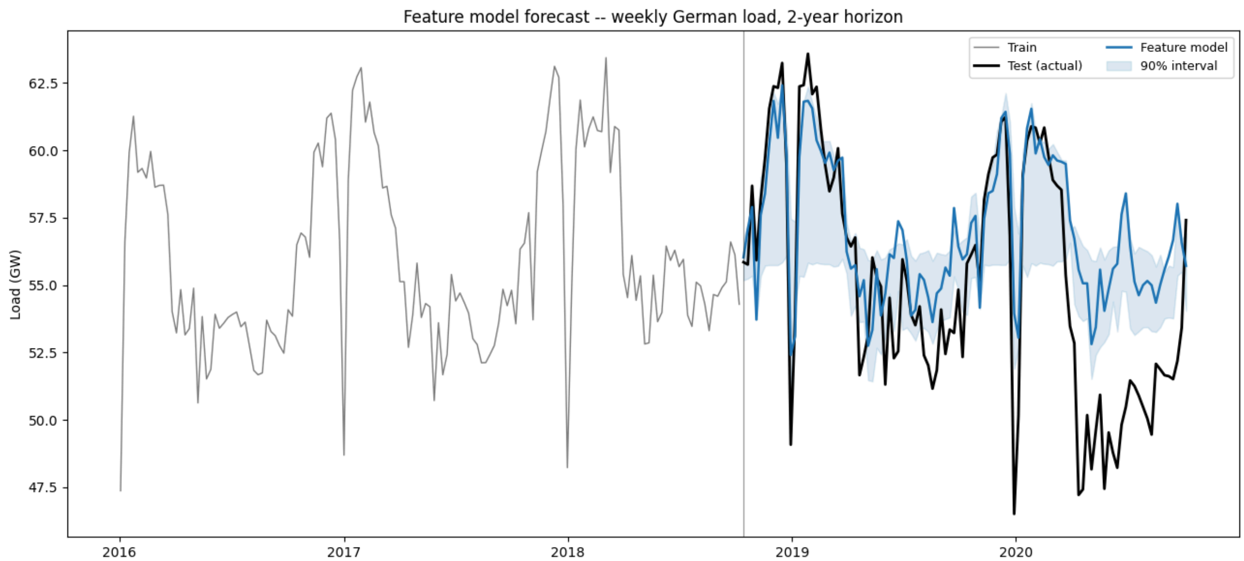

Figure 5: Feature model forecast

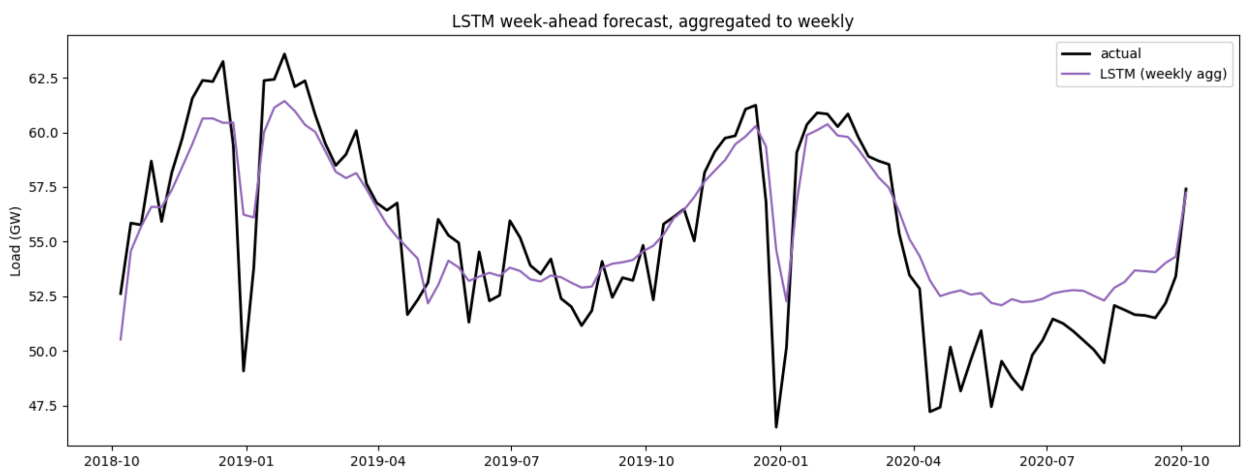

Figure 6: LSTM week-ahead forecast, aggregated to weekly

7. Error Analysis

Looking at the errors in the forecast in chronological order reveals the primary problem: errors were not particularly large or volatile over 2019, but were very large in the spring of 2020. This separation is precisely at the time of the COVID-19 structural break. This is because each model with a fixed origin overestimates the forecast (+1.7 to +2.8 GW bias) over this period, as it assumes a downward and upward trend relative to the fixed origin before and after 2020, respectively, when it was not expected that there would be a macroeconomic disruption in 2020.

The LSTM doesn't need to suffer from this divergence, as it continually adjusts to actuals. This was more apparent in the probabilistic evaluations of the SARIMAX model;

the coverage of the confidence interval for the model was approximately 51%, and the distribution of the residuals also had a high degree of non-normality, as evidenced by the small p-value in the Jarque-Bera test.

8. Discussion

The use of temperature covariates with the SARIMAX model was not consistent. In-sample, it gave an excellent fit of the model (AIC reduced from 301 to 290) and solved the remaining Ljung-Box lag-12 p-value failure (from 0.005 to 0.21). But out of sample, it failed to outperform that seasonal naive benchmark, and that huge +2.52 GW bias remained. Only heating_degree was significant (p=0.001), while cooling_degree was practically non-existent in Germany because the air conditioning load is negligible (p=0.88) and temp_mean was collinear. Furthermore, temperatures during a test period are technically "look-ahead" data, in that the weather that will occur during the test period cannot be forecast at the time the forecast is issued, and therefore are a conditional rather than an operational forecast. This study did not include holidays, which were not known from the calendar but instead represent an area of the predictive capability which was not captured.

Interpretability versus complexity is very simple. The SARIMAX model is very open to interpretation, including signed coefficients, principled confidence intervals, and formal diagnostics. The feature-based trees are in the centre and provide feature importances, but not directionality or inherent uncertainty. The LSTM is the highest complexity (requires extensive architecture, scaling, tuning and GPU compute) and the lowest interpretability (black box). Finally, there was no advantage of increased accuracy with increased complexity in this fixed-origin study.

The seasonal naive benchmark is the purest and most defensible recommendation for operational use for a 2-year-ahead planning horizon. It's played fairly, it doesn't need much upkeep, and it offers clear results. SARIMAX is a sensible improvement only when operators are very specific about wanting uncertainty intervals and/or explanations of weather impacts. However, on the other side of the equation, if the operational horizon is to be day/week ahead dispatch, a rolling SARIMAX or a rolling LSTM would be better, since their strength is with short-term re-anchoring.

9. Limitations and Future Work

A few limitations to this study exist. The geographical part of the temperature data didn't average over the whole country, but just over one place (Berlin). Holiday features were not incorporated and were very predictable sources of variation. Also, the change in structure was significant in 2020, and tree models were structurally limited in the ability to forecast outside of the training range, which is why they over-forecasted 2020. Finally, the uncertainty assessment of the methods (not the SARIMAX model) was very weak; the intervals for the tree and LSTM had an unreliable coverage of 36–54%. Future research should use a rolling-origin cross-validation to look at the robustness for each time horizon. Structural breaks can be handled using intervention analysis or dummy variables for COVID-19. The model could be enhanced by adding more probabilistic forecasting methods such as the Fourier terms for SARIMA to account for the longer seasonality (52 seasons) and/or adding more holiday covariates.

10. Conclusion

When it comes to the hard but meaningful 2-year-ahead fixed-origin forecasting challenge, there is no simple model that does a better job of beating the trivial seasonal naive model. The COVID-19 pandemic creates a structural break, such that the test error overwhelmingly dominates, which leaves the advanced models unable to perform as expected because they extrapolate from trends before the pandemic. For this long-term task, the increased complexity of machine learning and neural architectures was not warranted. Hence, the seasonal naive model is suggested for 2 years.

References

Kong et al. (2019) — Short-Term Residential Load Forecasting Based on LSTM RNN. *IEEE Trans. Smart Grid*, 10(1), 841–851. DOI: 10.1109/TSG.2017.2753802

https://doi.org/10.1109/TSG.2017.2753802

Free PDF:
https://www.researchgate.net/publication/319886687_Short-Term_Residential_Load_Forecasting_based_on_LSTM_Recurrent_Neural_Network

Bouktif et al. (2018) — Optimal Deep Learning LSTM Model for Electric Load Forecasting using Feature Selection and Genetic Algorithm: Comparison with Machine Learning Approaches. *Energies*, 11(7), 1636

https://doi.org/10.3390/en11071636

Open access: https://www.mdpi.com/1996-1073/11/7/1636

Open Power System Data (2020) — Time series data package

→ https://data.open-power-system-data.org/time_series/2020-10-06/